

Teja Ramyadevi A

DATA ENGINEER

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SUMMARY

Data Engineer with 4.5 years of experience designing and optimizing large-scale data solutions for healthcare and financial clients. Proficient in Azure and AWS cloud platforms, leveraging tools like Databricks, Redshift, and Glue for scalable ETL and streaming pipelines. Hands-on expertise in Apache Spark, Kafka, Hadoop, and Hive for processing high-volume structured and unstructured datasets. Strong programming skills in Python, SQL, and PySpark, with solid understanding of data modeling and governance. Experienced in creating interactive dashboards using Tableau and Power BI to deliver actionable business insights.

EXPERIENCE

Data Engineer | Cigna HealthCare, NJ

Mar 2024 – Present

- Designed and implemented ETL pipelines using Apache Spark for distributed data processing, enabling the ingestion and transformation of large-scale healthcare claims data with improved performance and scalability.
- Utilized AWS Redshift and S3 for data storage and retrieval, optimizing query performance and reducing latency in processing large datasets, achieving a 30% increase in data retrieval efficiency.
- Processed structured and unstructured data from SQL databases (PostgreSQL, MySQL) and external data sources (CSV, JSON) to ensure comprehensive claims data ingestion for analytics and reporting.
- Created interactive dashboards and visualizations using Tableau to provide stakeholders with actionable insights on claims processing, fraud detection, and operational metrics, facilitating data-driven decision-making.
- Used AWS EMR to set up streaming data pipelines with Apache Spark Streaming, processing real-time healthcare claim data as it arrived, enabling quicker detection of fraudulent claims and payment disputes.
- Developed data transformation scripts in Python to clean, preprocess, and aggregate large volumes of healthcare claims data from various sources, ensuring data consistency and quality for downstream analysis.
- Preprocessed and implemented features from structured healthcare claims data (claim amount, provider information, diagnosis codes) using Pandas and Scikit-learn, significantly improving model performance.
- Used ER diagrams to identify data dependencies and set up referential integrity constraints in the database, which improved the accuracy and consistency of claims data processing and reporting.
- Designed and implemented data marts based on user requirements, ensuring quick access to high-priority data subsets such as high-risk claims, payment disputes, and diagnosis trends, reducing query times by 30%.
- Utilized Apache Hive for querying and managing large-scale healthcare claims data stored in HDFS (Hadoop Distributed File System), enabling efficient analysis of structured and semi-structured data in a distributed environment.
- Employed AWS Glue Data Catalog as a central metadata repository to manage, store, and organize metadata from data sources (claims, providers, patients) which provided unified view of data and allowed for easier discovery and access.

Data Engineer | Accenture, India

Feb 2020 – Jul 2023

- Designed and implemented ETL pipelines using Apache NiFi and Python to ingest and transform financial transaction data, market data, and risk factors from multiple sources, including SQL Server, APIs, and third-party market data providers, ensuring efficient data flow and integration.
- Developed data models using Hive and HBase to store large-scale, high-velocity transactional data and risk indicators, optimizing query performance by 35% and supporting real-time analytics and fraud detection across diverse datasets.
- Architected and implemented cloud-based data solutions using Azure to store and process large-scale financial data, leveraging Azure Data Lake for unstructured data storage and Azure Blob Storage for scalable data ingestion.
- Enhanced fraud detection and risk assessment by integrating machine learning models into the data pipeline, reducing

false positives in compliance alerts by 15% and increasing the detection of high-risk transactions by 20%.

- Implemented machine learning models in PySpark using MLlib for predicting high-risk transactions and detecting fraud patterns, enhancing the ability to identify potentially fraudulent activities with 20% higher accuracy compared to traditional rule-based systems.
- Created dynamic reports in Power BI to track high-risk transactions, liquidity risks, and compliance issues, improving the accuracy of financial risk assessments and reducing manual reporting time by 30%.
- Designed MongoDB databases to store unstructured and semi-structured financial data, including transactional records, market data, risk indicators, enabling more flexible & scalable data storage solutions for large-scale datasets.
- Used Hadoop Distributed File System (HDFS) to store and manage large volumes of financial data across a distributed cluster, providing scalable and reliable storage solutions.
- Integrated Azure Synapse Analytics with Apache Kafka to ingest real-time financial transaction data and market data streams and enabled near-instantaneous analysis of transaction data, providing real-time risk detection and fraud alerts.
- Developed logging and error-handling mechanisms in Python to ensure reliable and transparent data processing, helping to track data pipeline performance and minimize downtime in the fraud detection system.
- Leveraged MapReduce for real-time fraud detection by processing incoming transaction data streams in parallel, reducing latency for flagging suspicious activities and triggering alerts in under 5 minutes.
- Designed and implemented data models in Snowflake for financial transactions, market data, and risk indicators, ensuring efficient storage and high-performance querying across large datasets with complex relationships.
- Developed visualizations in Matplotlib to display key financial risk metrics, such as transaction volumes, credit risk scores, and fraud detection rates, allowing stakeholders to quickly assess the health of financial portfolios.

SKILLS

Methodologies:	SDLC, Agile, Waterfall
Programming Language:	Python, R, SQL, Java, C++, C#, PowerShell (basic), .NET
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn
Visualization Tools:	Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP), Quick Sight
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook, IntelliJ
Database:	MySQL, PL/SQL, MSSQL, PostgreSQL, MongoDB, SQL Server
Data Engineering Concept:	Apache Spark, Apache Hadoop, Apache Kafka, Apache Beam, ETL/ELT, PySQL, PySpark
Cloud Platforms:	Microsoft Azure (Azure Blobs, Databricks, Data Lake), Amazon Web Services (AWS)
Other Technical Skills:	SSIS, SSRS, SSAS, Maven, Docker, Kubernetes, Jenkins, Terraform, Informatica, Talend, Snowflake, Google Big Query, Data Quality and Governance, Machine Learning Algorithms, Natural Language Processing, Big Data, Advance Analytics, Statistical Methods, Data Mining, Data Visualization, Data warehousing, Data transformation.
Version Control Tools:	Git, GitHub
Operating Systems:	Windows, Linux, Mac OS

EDUCATION

Master's in Data Analytics in Information Science from University at Albany - Albany, NY, USA.

Bachelor in Computer Science from **SCSVMV University**, India